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School of [School Name]

College of [College Name]

University of Lincoln

Submitted in partial fulfilment of the requirements for the Degree of [Your Degree Programme e.g. Bachelor of Science with Honours in Computer Science]

Supervisor: [Insert Supervisor Name Here]

[Month Year]

Acknowledgements

Write your acknowledgements here. You may wish to thank your supervisors, participants, friends, and family.

Abstract

Write your abstract here. Summarise the project, your approach, key findings, and conclusions.

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Figure 1 A picture of the Brayford from Google Images. 8

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Chapter 1

Introduction

This document is a project report template for the School of Computer Science at the University of Lincoln. It should give you some direction and instruction for formatting and presenting your project report. This template is designed for undergraduate project reports, however, templates are also available for postgraduate dissertations and doctoral theses on the same GitHub repository. If you have any suggestions or issues, please raise an issue on the GitHub repository.

This document has been generated using Typst, a modern typesetting system. The following subsections of this chapter demonstrate some basic Typst commands and formatting features. If you would prefer to use the supplied Word template, then you should aim to replicate the formatting shown here.

1.1 Subheading

1.1.1 Sub-subheading

1.2 Formatting Mathematics

Here are two equations:

$$a = b + 1$$

$$\frac{\hbar^2}{2m}\nabla^2\Psi + V(\boldsymbol{r})\Psi = -i\hbar\frac{\partial\Psi}{\partial t}$$

And here is some text with some nice inline maths, (x, y) wow γ so cool ρ.

1.3 Figures and Tables

Here is a sentence, and you can see a nice picture in Fig. 1.

Fig.1. A picture of the Brayford from Google Images.

Also, a table can be found in Table 1.

Table1. Here is a table. The caption goes above like this.

First Name	Last Name	Age
John	Doe	25
Jane	Smith	30
Alice	Johnson	22
Billy	Joel	28

1.4 Referencing

The included Harvard citation style file means that referencing will follow a Harvard style. You can cite references inline using the `@citekey` syntax. For example, here is a citation (Smith and Doe, 2024). You can also reference conference papers (Williams and Johnson, 2023) and books (Brown, 2022).

It is worth noting that the University of Lincoln's standard for referencing is Harvard, so you will not need to change this option. However, please double and triple check this to make sure you are using the correct referencing style. Also, ask your supervisor if you are unsure. This template uses a BibTeX (`.bib`) file for referencing. It is recommended to use a reference manager such as Zotero to export your references as a `.bib` file.

Chapter 2

Literature Review

The literature review is an essential requirement of any academic project. A comprehensive review of the literature will provide background to the project, and should be used to inform a set of requirements that your solution must meet. It also establishes that what you have done is the result of academic study, rather than an unfounded whim. You may use and build upon the literature review from your interim report, which you *must* reference. You should use the feedback from your supervisor to improve upon the final version.

It may be helpful to break up this chapter into sections, with each focused on a different topic or aspect of the project.

2.1 Aims and Objectives

Having situated your project within a body of relevant literature, you should now be in a position to state your aims and objectives. These should be broadly similar to those given in your proposal. Most projects will have one aim that is a broad statement of what the project will achieve. The objectives should be statements of how that aim will be achieved. Objectives should be Specific, Measurable, Assignable, Realistic and Time-related (SMART).

Chapter 3

Requirements Analysis

Drawing upon the research you have conducted, the goal of this chapter is to formalise a set of requirements that your project must fulfil in order to meet your objectives. This should expand upon the aims and objectives, stating what your solution must do to achieve the desired project outcomes. This chapter can be relatively short, and reference the aim(s) and objectives given at the end of the previous chapter.

Consider both functional and non-functional requirements, software/hardware environments, a risk analysis, and briefly how you will test and evaluate your artefact. For software-oriented projects, consider key stakeholders (end-users, business owners) when analysing the requirements of your software. This may be through primary research (i.e., interviews with potential end-users or clients) or secondary research (e.g., literature review, survey of existing products). For research-oriented projects, use your literature survey to highlight current gaps in knowledge that you hope to address. Consider what your artefact must do in order to obtain meaningful results.

Chapter 4

Design and Methodology

This section of the dissertation may vary significantly in both structure and content, depending on the type of project you are undertaking. However, it is expected that all projects have the following sections:

- Project management
- Risk analysis

The precise structure should be discussed with your supervisor, but some suggestions for additional sections are given below. The key thing to note here is that irrespective of the project type, you should justify the choices you have made, rather than simply choosing based on expediency or familiarity.

4.1 Software Development Projects

If the primary deliverable of your project is a software product, then you should consider subsections detailing your approaches to the following:

- Software development methodology (e.g., waterfall, scrum)
- Toolsets and machine environments (i.e., the software and hardware used)
- Design (e.g., UML diagrams, database schema, prototypes)
- Testing (i.e., the types of testing used)

This list is not exhaustive. For example, a games design project may include a game design document. However, it must be noted that if your project contains significant software development work, then most if not all of these sections should be present.

4.2 Research Not Involving Human Participants

For some projects, the main deliverables may come in the form of experimental results. For example, a project comparing several different algorithms may require little in the way of code, but require considerable experimentation and data analysis. As such, all methodological choices made should be documented here. Examples include:

- Dataset acquisition and annotation
- Algorithm/model design and selection
- Parameter tuning
- Performance metrics

Again, this list is not exhaustive, and you should still include relevant sections pertaining to the software artefact listed in the section above.

4.3 Research Involving Human Participants

For projects involving human participants, you will need to consider a hypothesis or research question that your project will answer. In addition to the sections outlined above, you may also need to provide details of:

- Participant recruitment
- Evidence that ethical procedures have been followed
- Study design (including hypotheses/research question as appropriate)
- Statistical analysis (i.e., how you will analyse the raw data)

If your study involves data collection by means of questionnaires, you may also wish to specify the questions here (or refer to them in an appendix).

Chapter 5

Implementation

5.1 Source Code Demo

This section should describe all components of your software implementation, emphasising particularly novel or nuanced components of code or its structure. There is no need to document every line of code, but you may include snippets as appropriate. You may wish to include code snippets and screenshots as figures to provide detail where necessary.

Chapter 6

Results and Discussion

This chapter should present the results of your work, be they in the form of a software product, experimental findings, or both. You should also link these results to your original objectives. For software-oriented projects, how well does it meet your original requirements? Provide data where possible, e.g., results of user testing, performance measures, etc. For research-oriented projects, you should present the data in an appropriate format (tables, charts, visualisations) and discuss what the data shows.

Chapter 7

Conclusion

This can be a relatively short chapter and is an opportunity to reflect on the project as a whole. It should broadly summarise the project, what has been achieved, its limitations, and the scope for future work. Discuss the development process and any lessons learned from undertaking an individual project.

References

Brown, R. (2022) *An Example Textbook*. Lincoln, UK: University Press.

Smith, J. and Doe, J. (2024) An Example Journal Article. *Journal of Example Research,* 42(3) 100–115.

Williams, B. and Johnson, S. (2023) An Example Conference Paper. In: *Proceedings of the International Conference on Examples.* 2023 50–62.

Appendices

This section is for any supplementary content that does not fit easily within the main dissertation. Examples include software documentation, user guides, large figures, logs of supervisor meetings, etc. This will not be marked directly but may be used to contextualise the main body of work.

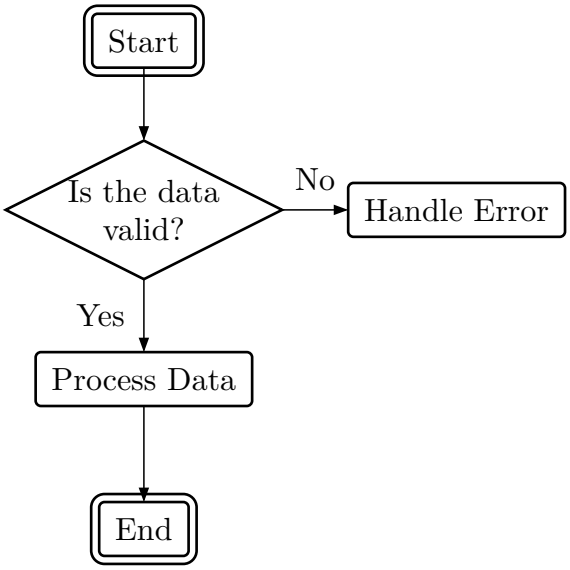
Appendix A: Code Snippets with Codelst

Here you can see a short snippet of code that was used to implement a simple Python sorting algorithm. The Codelst package provides syntax highlighting for code blocks:

```python
def bubble_sort(array):
    n = len(array)

    for i in range(n):

        already_sorted = True

        for j in range(n - i - 1):
            if array[j] > array[j + 1]:

                array[j], array[j + 1] = array[j + 1], array[j]

                already_sorted = False

        if already_sorted:
            break
    return array
```

Appendix B: Flowcharts with Fletcher

This uses the Fletcher package to create a flowchart. The package is available at: https://typst.app/universe/package/fletcher/



Word Count Summary

(excluding figures, tables, captions, and references)

Chapter	**Word Count**
Introduction	275
Literature Review	189
Requirements Analysis	164
Design and Methodology	367
Implementation	58
Results and Discussion	79
Conclusion	51
Total	**1183**

Overall Word Count: 1183 words